

FORM BLEND + DSPDS

A browser-based aim-form trainer and digital-spatial modeling system

Single-user, local-first, one-shot Codex build specification

PRIMARY OUTCOME	Transferable improvement in actual games.
CORE SYSTEMS	DSPDS spatial-model assessment, visible physical-movement baseline, and assisted Form Blend training.
BUILD PRIORITY	Low-latency input, credible results, useful coaching, strong UX, and minimal setup.
PLATFORM	Windows desktop browser; current Chrome or Edge; mouse and keyboard.
DATA MODEL	Local user/equipment profiles, immutable assessment history, evolving official model, JSON export/import.
VERSION	2.0 - August 3, 2026

Designed to minimize follow-up work: one large implementation pass, then only targeted tuning and defect repair.

Build decision summary

The project combines two complementary systems. DSPDS measures how visual spatial events are translated into unassisted physical mouse movement when camera and reticle feedback are removed. Form Blend then teaches a deliberate large-sweep -> stop -> smaller-flick -> micro-adjust sequence using the user's own movement signatures and spatial-model findings.

Final data-governance decision

Only formal DSPDS assessments update the official evolving spatial model. DSPDS practice sessions are fully saved and analyzed, but remain a separate dataset so feedback-assisted practice cannot contaminate formal assessment history.

Outcome priority order

Rank	Outcome	Design consequence
1	Transferable improvement in actual games	Every feature must justify how it improves closed-loop aiming or teaches transferable control.
2	Reduced jitter and corrective dependence	Analysis and coaching emphasize unnecessary reversals, high-frequency corrections, and late saves.
3	Better arm/wrist/fingertip blending	The physical baseline and Form Blend phase model remain central, but never claim anatomical certainty.
4	Understanding the internal spatial model	DSPDS produces a versioned, explainable sketch of visual-to-physical mapping over time.
5	Interesting analysis and visualization	Results should be visually legible and engaging, but never obscure the evidence.
6	Identifying directional weaknesses	All assessments retain direction, amplitude, reversal, and asymmetry data.
7	Better stopping control	Hard/soft stop mechanics and carry-through metrics remain core Form Blend components.
8	Finding an intuitive sensitivity range	Sensitivity is a useful DSPDS output, not the project's governing purpose or a single magic number.

Core architectural decisions

Area	Decision	Reason
Application	Local-first browser app	Fastest route to a polished 3D prototype without accounts or server infrastructure.
UI shell	React + TypeScript	Reduces one-shot implementation risk across many profile, calibration, review, and settings screens.
Real-time 3D	Three.js isolated from React rendering	The input/simulation loop remains deterministic and free of component-render timing.
Storage	IndexedDB + autosave + JSON import/export	Supports structured history and crash recovery without a backend.
Analysis	Transparent deterministic statistics	Small personal datasets do not justify opaque machine learning in version 1.
Design	Claude Design artifacts are visual authority	Codex implements an agreed design system instead of inventing UX during coding.
Input	Pointer Lock with unadjusted movement request	Captures relative motion and separates raw samples from rendered camera movement.

Release rejection criteria

- Poor UX/UI: the app feels cheap, confusing, visually inconsistent, or difficult to navigate.
- Laggy input: camera response, pointer lock, or sample capture feels delayed or unstable.
- Inconsistent results: equivalent repeat assessments contradict each other without a visible confidence or reliability explanation.
- Arbitrary conclusions: insights cannot be traced to named metrics, trial groups, and stored raw data.
- Low practical value: the app produces data but no beneficial knowledge, coaching, or observable form gain.
- Excessive setup: repeated configuration takes precedence over training.

Document map

Section	Purpose
1. Product definition	Defines the training thesis, success standard, and boundaries.
2. Profile and storage	Defines users, equipment contexts, history, recovery, and export.
3. Input and view setup	Defines DPI, sensitivity, cm/360 testing, FOV, and viewpoint behavior.
4-6. DSPDS	Defines governance, all five scenarios, analysis, and the evolving spatial model.
7. First-use flow	Creates the shortest route from launch to useful information.
8. Physical baseline	Captures arm-, wrist-, fingertip-, and freestyle-labeled visible movement signatures.
9-10. Form Blend	Defines the guided exercise, anti-crutch rules, scoring, and coaching.
11. UI/UX	Defines Claude Design deliverables and active-training visual rules.
12. Technical architecture	Defines repository, loops, schemas, algorithms, and persistence.
13-15. Execution	Defines milestones, tests, tuning, and a one-shot Codex handoff.
16. Scope creep	Lists additions with promising ROI after the base system proves value.

1. Product definition

Form Blend + DSPDS is a single-user 3D aim-training application designed to improve real-game mouse control by connecting three layers: open-loop spatial intuition, observable closed-loop mouse behavior, and a guided blended-movement exercise. It assumes the user participates in good faith when asked to isolate arm, wrist, fingertip, or freestyle movement.

Training thesis

Layer	Question answered	System
Open-loop spatial mapping	How much and how does the user naturally move the mouse for a visible spatial event when input cannot change the view?	DSPDS formal assessment and practice.
Closed-loop physical signatures	What do user-labeled arm, wrist, fingertip, and freestyle movements look like during ordinary visible aim?	Four-part physical movement baseline.
Guided blending	Can the user transport broadly, remove momentum, correct at a smaller scale, and finish precisely without brute-forcing the target?	Form Blend assisted and unassisted training.

Compound movement taught by Form Blend

ARM-SCALE TRANSPORT	->	STOP / TRANSITION	->	WRIST-SCALE CORRECTION	->	FINGERTIP MICRO	->	CLICK
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What the application may and may not infer

Good-faith personal model

Mouse telemetry cannot prove which anatomical structure moved the device. The labels arm, wrist, fingertip, and freestyle are supplied by the user during calibration. Future segments may be described as arm-like, wrist-like, or fingertip-like only when similarity is supported; the interface must never present this as visual proof of muscle use.

Definition of benefit

A useful first version should produce evidence of change after several sessions, not merely a compelling dashboard. The primary comparison is unassisted Form Blend performance against the user's original baseline and early sessions.

Evidence family	Desired movement
Jitter and correction dependence	Fewer high-frequency reversals, smaller late corrections, and less correction distance after first arrival.
Movement blending	Clearer scale separation between primary transport, secondary correction, and micro-adjustment.
Stopping	Lower carry-through and shorter settle time without sacrificing arrival accuracy.
Transfer	Equal or improved visible-aim accuracy and timing in closed-loop validation scenarios.
Reliability	Repeated assessments show stable central estimates or honestly communicate low confidence.

Version 1 non-goals

Excluded	Reason
Accounts, login, cloud sync, subscriptions	No benefit to proving the core training and analysis model.

Excluded	Reason
Multiplayer, leaderboards, achievements	They encourage score farming before form validity is established.
Opaque machine-learning classifier	The personal dataset is small; decisions must remain inspectable and debuggable.
Weapon simulation, recoil, movement combat	Adds game complexity unrelated to the initial motor-learning proof.
Skeletal body tracking	The first version trusts isolation instructions and measures observable mouse behavior.
Mobile, controller, Safari, Firefox guarantees	The initial platform is Windows desktop Chrome/Edge with mouse and keyboard.

2. User profile, equipment contexts, and storage

The first build supports one local user profile, but its data model must permit multiple users later. Each mouse, DPI, sensitivity, FOV, or view configuration lives inside an equipment profile so hardware changes are not confused with personal improvement.

Profile hierarchy

```

UserProfile
  id, displayName, createdAt, schemaVersion
  preferences
  equipmentProfiles[]

EquipmentProfile
  mouseName, dpi, handedness
  selectedCmPer360, measuredCmPer360
  countsPerCm, inputVerificationState
  verticalFovDeg, horizontalFovDeg, viewMode
  dspdsAssessments[]
  dspdsPracticeSessions[]
  spatialModelSnapshots[]
  physicalBaselines[]
  formBlendSessions[]
  pinnedSessionIds[]

```

Storage behavior

Requirement	Behavior
Primary store	IndexedDB with explicit schema versioning and migration functions.
Autosave	Persist after every completed trial and at every state transition that would be painful to repeat.
Crash/refresh recovery	Resume the incomplete block from the last committed trial; never merge partial raw samples into a valid trial.
Persistent storage	Request browser persistent storage where supported and show whether it was granted.
Export/import	Full JSON export, summary-only JSON export, and verified import with schema/version checks.
Offline behavior	No internet dependency after the app bundle is loaded.
Deletion	No silent deletion. Warn before storage cleanup and provide export first.

Retention policy

- Keep all formal assessment summaries and all spatial-model snapshots indefinitely.
- Keep all session summaries indefinitely unless the user explicitly deletes them.
- Keep raw event-level telemetry for at least the latest 100 sessions.
- Allow any session to be pinned so its raw data is never automatically removed.
- Include raw data in a full export when present; identify sessions whose raw data has been pruned.

Formal assessment governance

Dataset	May update official DSPDS model?	Feedback during run?	History
Formal DSPDS assessment	Yes	No trial-level correction feedback	Immutable assessment record plus new model snapshot.
DSPDS practice	No	Allowed after movement or block	Fully archived in a separate practice dataset.
Closed-loop validation	No direct update	Blinded until block completion	Compared with DSPDS recommendations and stored separately.
Physical baseline	No direct DSPDS update	Block-level summary only	Creates labeled visible-movement signatures.

When recalibration is recommended

- Mouse or sensor change.
- DPI change.
- Selected sensitivity change greater than 10 percent.
- Major grip, posture, desk, or arm-support change.
- FOV change when comparing visually sensitive metrics.
- Long period of inactivity.
- User requests a fresh baseline or formal DSPDS reassessment.

Recalibration is a recommendation, not a training block. The app must never prevent the user from practicing because a profile is unverified.

3. Input, sensitivity, FOV, and viewpoint setup

Input profile states

State	Available information	Permitted output
Relative only	Raw input counts; no trusted DPI or physical measurement	Directional ratios, count consistency, relative gain, and spatial-model features.
Calculated	DPI plus selected cm/360	Approximate physical movement and sensitivity outputs, labeled unverified.
Physically verified	DPI and completed physical cm/360 tester	Verified counts-per-cm and physically expressed DSPDS sensitivity ranges.

Sensitivity representation

The engine stores camera gain as degrees per raw input count. FOV never changes this physical rotation mapping.

```
countsPer360 = dpi * (cmPer360 / 2.54)
yawDegPerCount = 360 / countsPer360

// Example: 1600 DPI at 40 cm/360
countsPer360 ~= 25,197
yawDegPerCount ~= 0.01429
```

Physical cm/360 tester

The tester is optional and accessible from initial setup, Home, Pause, Equipment Profile, and any DSPDS result whose physical confidence would improve after verification.

1. Launch a first-person calibration scene with a distinct vertical reference line.
2. Ask the user to mark the physical position of the mouse's right edge.
3. First left click begins raw input capture.
4. The user turns exactly 360 degrees until the reference line realigns.
5. Second click ends the run; the app records accumulated raw counts.

6. The user marks the new right-edge position, measures the distance, and enters centimeters.
7. Repeat three times. Report the median distance, variation, counts-per-cm, and estimated effective DPI.

```
countsPerCm = median(abs(horizontalCounts_i) / enteredDistanceCm_i)
estimatedEffectiveDpi = countsPerCm * 2.54
measuredCmPer360 = median(enteredDistanceCm_i)
```

Source-of-truth rule

The physical test verifies the profile but does not silently change selected sensitivity. The user must approve any recommended profile update. Changing sensitivity during a formal baseline starts a new comparison context.

FOV behavior

- Expose both vertical FOV and horizontal 16:9 FOV; clearly indicate which field is being edited.
- Store one canonical vertical-FOV value internally and derive horizontal FOV from aspect ratio.
- Initial allowed vertical range: 60 to 120 degrees.
- Target geometry uses angular or world-space coordinates, never screen-pixel distance.
- Changing FOV changes apparent motion and target size but never degrees per mouse count or cm/360.

```
horizontalFov = 2 * atan(tan(verticalFov / 2) * aspectRatio)
```

First- and third-person behavior

Mode	Version 1 behavior	Comparison rule
First person	Canonical DSPDS and calibration view. Camera rotates at the aim pivot.	Primary assessment context.
Third person	Simple capsule/ovoid, centered trailing camera, no locomotion or animation, same angular input model.	Form Blend results stored separately from first-person sessions.

DSPDS always runs in first person in the initial release because third-person offset and parallax would create a separate spatial task.

Raw input requirements

- Use Pointer Lock and request unadjusted movement when available.
- Capture every mouse event independently of the render loop into a preallocated ring buffer.
- Never infer event timing from frame timing.
- Record raw deltas even when an assisted stop freezes or damps the displayed camera.
- Pointer-lock loss pauses the current trial and marks any incomplete sample invalid.
- Display an input diagnostic: raw-input request status, event cadence, frame cadence, dropped-sample warnings, and effective gain.

4. DSPDS definition and assessment rules

DSPDS stands for Digital-Spatial Predilection Detection System. It observes how the user's hand responds to visible spatial events when the usual closed-loop relationship between physical mouse movement and camera/reticle movement is intentionally removed.

DSPDS purpose

- Measure the internal mapping between visual angular displacement and physical mouse movement.
- Create a rough, evolving sketch of the user's internal spatial-reasoning model for aim.
- Estimate intuitive sensitivity ranges without presenting a single perfect number.
- Identify directional, amplitude-dependent, reversal, timing, and consistency discrepancies.
- Compare open-loop spatial intuition with later closed-loop visible aiming performance.
- Provide data that can shape future exercises targeting mind-body and spatial-intuition weaknesses.

Shared visual rule

No blind-screen test

The environment and target/path remain visible. No reticle or cursor appears. User input produces no camera or target response

during the measured movement. The screen never abruptly goes black. The application may reposition or automatically rotate the camera only between or as part of defined visual events.

Assessment and practice separation

Mode	Purpose	Feedback	Model effect
Formal assessment	Comparable snapshot of current open-loop mapping	Results hidden until each scenario block ends	Creates a new immutable assessment and updates the official model.
Practice	Explore or train the same spatial-reflex tasks	Block or trial feedback allowed	Saved separately; never silently updates the official model.

Shared trial lifecycle

READY	->	VISUAL EVENT	->	SOFT START CUE	->	PHYSICAL RESPONSE	->	SETTLE / CLICK	->	SAVE
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Movement completion modes

Mode	Behavior
Automatic	End after minimum movement plus 120-180 ms below a velocity threshold.
Click to complete	End on left click after minimum movement and minimum duration.
Automatic with click failsafe - default	Use settle detection normally; allow click to end a movement that does not settle cleanly.

Assessment duration

A complete formal DSPDS assessment should target 10-12 minutes. Scenario blocks may be run separately in practice. The formal protocol should display estimated remaining time and permit safe pause/resume between blocks, never during an active sample.

Assessment contamination controls

- Hide endpoint error, sensitivity estimate, and raw-path overlays until the block ends.
- Randomize directions and amplitudes with a reproducible seed.
- Do not display numeric target angle during collection.
- Provide neutral recenter periods that ignore input when the mouse approaches the edge of the usable pad.
- Do not immediately repeat a failed trial with an identical visible sequence unless the failure was technical.
- Preserve completion mode and any failsafe click in the trial record.

5. DSPDS scenario specifications

5.1 Single-target displacement

A flat wall fills the forward view. One constant-size sphere appears at a random angular offset from screen center. The camera remains fixed and no reticle is shown. The user moves the mouse the amount and direction that feels naturally sufficient to reach the visible sphere. The target is treated as acquired regardless of the actual response.

Parameter	Initial specification
Origin	Every formal trial begins at screen center.
Angular tiers	2, 4, 8, 12, 18, 27, and 40 degrees. Enable 50 degrees only when horizontal FOV leaves safe margin.
Direction weighting	Horizontal and diagonals receive more samples; vertical directions remain represented.

Parameter	Initial specification
Target size	Constant angular radius within the assessment.
End condition	Selected completion mode.
Primary outputs	Projected displacement, orthogonal error, direction error, implied gain, speed profile, smoothness, stopping tail, and consistency.

5.2 Chained target flicking

The view always contains a current target and a next target. The user physically flicks toward the current target with no camera response. At movement completion, the application rotates the camera to center the intended current target regardless of the user's physical endpoint. The next target becomes current and another target is generated ahead.

Rule	Behavior
Chain length	5-8 targets per chain.
Camera transition	Short smooth auto-rotation, initially 80-120 ms.
Visibility	Current and next target remain visible; only the active vector is measured.
Sequence design	Same-direction continuation, partial reversal, full reversal, diagonal transition, large-to-small, and small-to-large.
Why it matters	Preserves a changing frame of reference while preventing visual correction inside each flick.
Primary outputs	Gain drift through the chain, reversal asymmetry, amplitude adaptation, accumulated bias, and consistency after camera recentering.

5.3 Predictable camera-follow tracking

A sphere strafes through a predictable path while the application automatically rotates the camera to keep it approximately centered. The environment pans, but the user's mouse input cannot affect the view. The user physically mirrors the camera movement as though causing it.

Parameter	Initial specification
Paths	Left-right-left, right-left-right, slow diagonal sweep, and shallow vertical variant.
Motion blocks	Smooth sinusoidal reversal and constant-speed motion with firmer turnaround.
Cycles	2.5 cycles per trial, ending near center.
Target behavior	Target remains near center; no input-driven reticle.
Primary outputs	Best-fit gain, response lag, phase error, accumulated drift, reversal timing, speed coupling, and left/right asymmetry.

5.4 Dynamic target-follow tracking

Dynamic tracking is implemented as two distinct tests because they represent different internal models.

Variant	Visual task	User interpretation
A. Static camera	Sphere moves around the screen while the camera stays fixed.	Move an invisible reticle with the hand.
B. Automatic tracking camera	Camera follows the sphere so it remains near center and the environment moves.	Mirror the camera motion physically.

- Use smooth readable seeded paths with bounded acceleration.
- No teleportation, reactive evasion, or abrupt un-signalized 180-degree reversal.
- Use gradual speed changes and a flat angular path in version 1.
- Do not include depth changes until the 2D angular model is stable.

- Record gain, lag, phase, directional error, response to speed change, and prediction at curvature changes.

5.5 Progressive straight-line tracing

A straight line grows from screen center at a known constant speed. The user moves the mouse as though tracing the line while no reticle, cursor, or camera response reveals the physical path.

Parameter	Initial specification
Directions	Horizontal, vertical, and diagonal.
Lengths	Short, medium, and long angular equivalents.
Presentation speeds	Two speeds in the formal protocol.
Path shape	Straight only in version 1.
Recenter	Only when the mouse approaches the physical edge; recenter input is excluded.
Primary outputs	Direction error, length ratio, straightness, speed coupling, lag, and endpoint consistency.

Scenario implementation gate

First usable DSPDS gate

Single-target displacement, chained target flicking, and predictable camera-follow tracking must work and produce inspectable outputs before dynamic tracking and line tracing are added. All five remain mandatory for the complete initial release.

6. DSPDS analysis and evolving spatial model

No opaque model in version 1

DSPDS uses deterministic vector math, robust summaries, and named comparisons. A future learned model may be useful after many formal assessments exist, but the first release must make every conclusion reproducible from stored trials.

Single-displacement inference

For an intended angular vector and a physical response vector, project the response onto the intended direction. The parallel component estimates intuitive gain; the orthogonal component measures directional discrepancy.

```
thetaDeg = magnitude(intendedAngularVector)
parallelCounts = dot(rawCountVector, unit(intendedAngularVector))
orthogonalCounts = cross2D(rawCountVector, unit(intendedAngularVector))

relativeGainCountsPerDeg = parallelCounts / thetaDeg

if countsPerCm is verified:
    projectedCm = parallelCounts / countsPerCm
    impliedCmPer360 = projectedCm * (360 / thetaDeg)
```

Tracking inference

For predictable and dynamic tracking, compare the known automatic angular-motion signal with the raw mouse-velocity signal. Search a small lag window and solve for the gain that best maps input to the visual motion.

```
for lagMs in -250..250:
    align mouseVelocity(t - lagMs) with idealAngularVelocity(t)
    gain = argmin_g sum(||g * mouseVelocity - idealAngularVelocity||^2)
    error = normalizedResidual(g, lagMs)
choose lagMs and gain with lowest robust error
```

Report fitted gain, lag, phase error, drift, reversal timing, and confidence. Do not interpret reaction delay as sensitivity by itself.

Line-tracing inference

- Integrate raw samples into a physical 2D path.
- Normalize time so the displayed growing line and mouse path can be compared despite small start delay.

- Measure line-angle error, response-length ratio, orthogonal deviation, path straightness, speed correlation, lag, and endpoint stability.
- Keep direction and speed conditions separate before forming an overall summary.

Amplitude bands

Band	Single-target angles	Primary interpretation
Micro	2-4 degrees	Fine internal scale and small-movement consistency.
Small	8-12 degrees	Fingertip/wrist transition region and direction bias.
Medium	18-27 degrees	Wrist/arm transition region and gain stability.
Large	40 degrees and optional 50 degrees	Broad transport intuition and large-amplitude compression/expansion.

Official spatial-model contents

Model component	Stored result
Intuitive gain	Relative counts/degree and verified implied cm/360 where possible.
Scale behavior	Separate micro, small, medium, and large estimates and cross-scale compression/expansion.
Direction map	Horizontal, vertical, diagonal, left/right, up/down, and reversal summaries.
Consistency	Within-condition median absolute deviation and trial-to-trial endpoint spread.
Timing	Response lag, time to peak, stopping tail, and reversal timing.
Path quality	Straightness, smoothness, high-frequency reversal rate, and orthogonal error.
Tracking model	Fitted gain, phase error, accumulated drift, and response to speed changes.
Reliability	Valid sample count, dispersion, technical flags, completion mode effects, and confidence.

Model snapshot and evolution

1. Each formal assessment creates an immutable assessment result.
2. Each result creates a SpatialModelSnapshot with scenario-level and combined estimates.
3. The current official model uses a robust recency-weighted summary of up to the three latest valid formal assessments.
4. The latest assessment and the smoothed official model are shown separately so change is not hidden by averaging.
5. All prior snapshots remain available as a timeline; practice is shown alongside but never included in the official aggregate.

```
officialCurrent = weightedRobustAggregate(
  lastValidFormalSnapshots(max = 3),
  recencyWeights = [1.0, 1.5, 2.0]
)
```

Confidence model

Factor	Raises confidence	Lowers confidence
Sample sufficiency	Enough valid trials in every required condition	Missing tiers, repeated invalid trials, or early exit.
Within-block dispersion	Stable gain and direction within matching trials	Large unexplained spread.
Cross-scenario agreement	Displacement, chain, and tracking estimates broadly agree	Scenarios imply incompatible gains.

Factor	Raises confidence	Lowers confidence
Technical quality	Stable input cadence and no pointer-lock interruption	Dropped events, cadence anomalies, or frequent click failsafe.
Test-retest stability	Recent formal assessments show consistent structure	Large changes without profile or performance explanation.

Sensitivity output

DSPDS never claims one ideal sensitivity. It reports a preferred range and explains how the estimate changes by amplitude and direction.

```
Intuitive range: 34-39 cm/360
Central estimate: 37 cm/360
Micro estimate: 40 cm/360
Medium estimate: 37 cm/360
Large estimate: 34 cm/360
Directional note: rightward flicks imply lower gain than leftward flicks
Confidence: Moderate
```

Insight traceability

No mysterious result

Every strength, weakness, or recommendation must link to the contributing scenario, condition, sample count, median, dispersion, and confidence. The user must be able to open a replay or raw-path view after the block.

7. First-launch and recurring user flows

First-launch objective

The first session must reach useful information quickly. Physical cm/360 verification is encouraged but optional. DSPDS occurs before the four-part physical baseline because its open-loop findings can help choose movement amplitudes and provide context for later visible aiming.

First-session flow

1. Landing: Begin setup or continue recovered session.
2. Create the single local user and one equipment profile.
3. Enter mouse name, dominant hand, optional DPI, selected cm/360, and FOV.
4. Run a 20-second input diagnostic: pointer lock, movement, click, Escape recovery, event cadence, and frame cadence.
5. Offer the physical cm/360 tester; allow Skip for now.
6. Run formal DSPDS blocks in the defined order and create the first spatial-model snapshot.
7. Run the four-part visible physical baseline using equivalent target sets.
8. Show one compact combined profile: spatial model, movement signatures, reliability, and no more than three useful observations.
9. Demonstrate Form Blend twice and begin the guided onboarding sequence.
10. End with assisted versus unassisted comparison and one next-session focus.

Home screen

Primary action	Behavior
Continue Form Blend	Resumes the current recommended training progression.
DSPDS assessment	Starts or schedules a formal snapshot; shows last assessment age and confidence.
DSPDS practice	Opens scenario practice without affecting the official model.
Physical baseline	Runs arm/wrist/fingertip/freestyle recalibration.

Primary action	Behavior
Profile insights	Shows current model, movement signatures, trends, and evidence links.
Input and view setup	DPI, sensitivity, FOV, viewpoint, diagnostics, and cm/360 tester.

Pause menu

- Resume, restart current trial, exit after saving, diagnostics, sensitivity/FOV display, cm/360 tester entry, and DSPDS completion-mode toggle.
- Changing sensitivity or FOV during an assessment requires explicit confirmation and ends the current comparison block.
- Pause input is excluded from all analysis.

8. Four-part physical movement baseline

This baseline uses ordinary visible aiming. It captures the user's good-faith arm-only, wrist-only, fingertip-only, and freestyle signatures after the initial DSPDS assessment. The same seeded target vectors are used in shuffled order for every block.

Isolation instructions

Block	On-screen instruction	Purpose
Arm only	Move primarily with your arm. Keep the wrist and fingers relatively quiet.	Broad transport signature.
Wrist only	Plant the forearm and move primarily at the wrist.	Medium-scale correction signature.
Fingertip only	Keep the heel of the hand and wrist planted. Move primarily with your fingertips.	Micro-adjustment signature.
Freestyle	Use whatever combination feels natural.	Natural blending and transition signature.

Protocol

Block	Initial valid samples	Initial angular envelope	Presentation
Arm	16	18-32 degrees, adjusted using DSPDS large/medium range	Generous target, no time pressure.
Wrist	16	6-14 degrees, adjusted using DSPDS small/medium range	Medium target.
Fingertip	20	0.8-5 degrees, adjusted using DSPDS micro/small range	Small target, magnified review only after block.
Freestyle	24	Mixed amplitudes matching the same seed families	Ordinary visible aiming.

Per-trial sequence

CENTER	->	READY CUE	->	TARGET	->	VISIBLE AIM	->	ARRIVAL / CLICK	->	RETURN
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Validity and contamination rules

Condition	Response
Premature movement before target onset	Repeat without penalty.
Wrong-direction initial launch	Mark invalid and repeat later with a different target.
Pointer-lock loss or cadence defect	Technical invalidation; save diagnostic context.
Movement far outside the intended scale	Keep as labeled data but lower signature confidence; do not accuse the user of using the wrong body part.

Condition	Response
Crawled attempt	Preserve sample, label as cautious/low-speed, and exclude only if below minimum meaningful movement.

Movement features

```

MovementFeatures {
  angularDistanceDeg
  durationMs
  peakSpeedDegPerSec
  timeToPeakMs
  primarySubmovementRatio
  pathEfficiency
  firstArrivalErrorDeg
  overshootDeg
  correctionDistanceDeg
  correctionCount
  directionReversalCount
  settleTimeMs
  velocityAtClick
  highFrequencyReversalRate
  speedCv
  sparclikeSmoothness
}

```

Signature construction

1. Compute the same named feature vector for every valid labeled trial.
2. Store medians and median absolute deviations by block, direction family, and amplitude family.
3. Normalize future Form Blend segments against the appropriate scale and direction context.
4. Use feature-distance similarity only as a coaching aid; amplitude and phase order remain the primary constraints.
5. Display similarity as High, Moderate, Low, or Unresolved rather than false anatomical certainty.

9. Core Form Blend exercise

Every repetition contains one linked movement problem. The user transports broadly into a gate, removes momentum, performs a smaller correction, reveals or acquires a micro target, and clicks under control.

Target construction

Element	Function	Visual treatment
Transport gate	Defines the end region for the broad primary sweep.	Large translucent ring or sphere segment.
Transition anchor	Defines where momentum must be removed.	Compression ring that brightens as velocity falls.
Secondary target	Requires a smaller distinct flick after stability.	Crisp sphere or disc offset from the anchor.
Micro target	Requires final small correction and controlled click.	Small luminous core revealed near first arrival.

State machine

```

READY
-> PRIMARY_SWEEP
-> TRANSITION_CAPTURE
-> STABILITY_HOLD
-> SECONDARY_FLICK
-> MICRO_ADJUST
-> CLICK
-> REP_FEEDBACK
-> RESET

```

Phase requirements

Phase	Completion condition	Primary evaluation
Primary sweep	Enter transport gate.	Decisive transport, efficient path, limited early corrections, scale appropriateness.
Transition	Velocity remains below threshold for stability window.	Carry-through, stop latency, reversal after stop, and pre-stop deceleration.
Secondary flick	Enter correction target after stability.	One clear smaller submovement, overshoot, path, and wrist-like similarity.
Micro-adjust	Acquire micro target.	Correction count, jitter, settle time, and fingertip-like similarity.
Click	Click inside target while valid.	Accuracy, velocity at click, and premature-click state.

Demonstration sequence

1. Show the full repetition at normal speed with labeled movement roles.
2. Replay at half speed with broad path, transition compression, smaller correction, and magnified micro finish.
3. Explain that the application measures movement structure and similarity, not anatomical certainty.
4. Show the carry-through concept: the displayed camera can stop while raw input continues to be recorded.
5. Begin the first assisted repetition immediately; do not send the user back through menus.

Assisted onboarding progression

Stage	Typical reps	Guidance
Full demonstration + hard stop	1-3	Ghost structure, phase labels, displayed camera freezes at transition.
Hard stop	4-6	No ghost; show carry-through after the rep.
Reduced hard stop	7-10	Shorter catch and fewer labels.
Soft damping	11-14	Camera gain decreases through a transition zone.
Cue only	15-17	Audio/visual transition cue without physical assistance.
Unassisted	18-20	Full evaluation and comparison.

Progression is competency-based with a maximum tutorial length. Repeated clean control may advance sooner; persistent failure can trigger one additional demonstration without creating permanent assistance.

Hard-stop mechanic

```
on enterTransitionBoundary:
    catchOrientation = displayedCamera.orientation
    state = TRANSITION_CAPTURE

while state == TRANSITION_CAPTURE:
    displayedCamera.orientation = catchOrientation
    rawCarryThrough += magnitude(rawMouseDelta)
    appendRawSample(rawMouseDelta)
    if rawSpeed < stopThreshold for stabilityWindowMs:
        releaseToSecondaryPhase()
```

Soft-stop mechanic

```
progress = clamp(distanceIntoDampingZone / zoneDepth, 0, 1)
displayGain = lerp(1.0, minimumGain, smoothstep(progress))
displayedDelta = rawDelta * displayGain
// Always analyze rawDelta, never displayedDelta.
```

DSPDS-informed generation

- Weight more trials toward directions with unstable or biased internal mapping.

- Sample amplitudes near transitions between micro, small, medium, and large spatial-model bands.
- Use DSPDS gain as context for geometry, never as a silent sensitivity change.
- Keep scenario seeds reproducible and record the generator version.
- Preserve balanced exposure so adaptive targeting cannot hide stronger directions.

10. Form validity, scoring, coaching, and anti-crutch rules

A hit is not automatically a valid repetition

Violation	Result
Primary gate never entered	Incomplete repetition.
Secondary target attempted before stability	Form invalid even if the final target is hit.
Entire sequence performed as one uninterrupted ballistic flick	Form invalid: phase separation missing.
Extremely slow drag through every phase	Form invalid or capped: cautious drag.
Click before micro target is valid	Premature click and repetition failure.
Repeated sawtooth corrections	Allow completion, but correction-dependence limits phase quality.

Weakest-phase result

```
repQuality = min(
  primarySweepQuality,
  transitionQuality,
  secondaryFlickQuality,
  microControlQuality
)
```

Accuracy is always displayed separately. A fast hit cannot erase poor stopping or repeated corrections.

Phase scoring

```
phaseScore = 100
- pathPenalty
- timingPenalty
- overshootPenalty
- correctionPenalty
- jitterPenalty
- invalidStatePenalty
```

Feedback hierarchy

1. Major validity error that prevents the intended exercise.
2. Jitter or correction dependence that most damaged control.
3. Movement-scale blending mismatch.
4. Stopping/carry-through issue.
5. Directional weakness relevant to the current rep.
6. Sensitivity observation only when repeated and confidence-supported.

Displayed feedback format

```
Successful hit

Primary sweep: Clean
Transition: Carried through
Secondary flick: Clean
Micro-adjustment: Rushed

Next rep:
Begin removing speed slightly earlier. Do not aim the broad sweep at the final dot.
```

Coaching constraints

- Give one primary next-repetition cue, not a data dump.
- Use plain language first; allow deeper evidence to be expanded afterward.

- Never claim a body part was used with certainty.
- Do not change sensitivity or assistance without explicit visible notice.
- Do not reward brute-force speed outside the user's valid movement window.
- Retain accuracy, speed, and form as separate concepts.

11. UI/UX and Claude Design handoff

Design direction

Claude Design should produce a hybrid visual system: an OpenAim-like restrained 3D training space combined with Fabled Aim's softer, organic, readable analysis. Active exercises remain nearly free of chrome; analysis gains depth only after the movement is complete.

Visual principles

Principle	Requirement
Low cognitive load	One primary action per screen; no dense metric walls during training.
Spatial clarity	Targets, gates, paths, and camera motion remain immediately understandable.
Evidence without intimidation	Plain-language conclusion first, expandable metric and replay detail second.
Continuity	Home, DSPDS, baseline, Form Blend, and results feel like one product.
Performance first	No visual effect may compromise input response or stable frame timing.
No cheap prototype feel	Typography, spacing, motion, empty states, and loading/recovery states are fully designed.

Required Claude Design screens

- First launch and profile creation.
- Home and continue-training state.
- Equipment profile, sensitivity/FOV setup, and input diagnostics.
- Physical cm/360 tester instructions and results.
- DSPDS assessment/practice selection and completion-mode control.
- Instruction and active state for each of the five DSPDS scenarios.
- DSPDS block results, official spatial-model summary, history, and evidence detail.
- Four-part physical baseline instructions, active HUD, and combined signature summary.
- Form Blend demonstration, assisted HUD, unassisted HUD, and repetition feedback.
- Pause, recovery, import/export, storage warning, and failure states.
- Session review, trend view, raw-path replay, and profile insights.

Active-training HUD

Visible	Hidden until later
Target/environment, minimal phase cue, optional rep count, pause affordance	Live graphs, sensitivity estimates, detailed scores, endpoint overlays, and correction advice during movement.

Movement visual language

Role	Suggested treatment	Meaning
Primary transport	Broad cool arc or translucent gate	Commitment and movement scale.
Transition	White compression ring or subtle contraction	Momentum removal.
Secondary correction	Crisp solid target	Smaller distinct submovement.
Micro finish	Small luminous core and post-rep magnification	Fine control.
DSPDS intended path	Subtle neutral geometry	Visible spatial event, not a score target.

Role	Suggested treatment	Meaning
Reconstructed response	Shown only in results/replay	Physical movement interpretation.

Design-to-code authority

Source-of-truth hierarchy

This specification governs function, measurement, data, milestones, and acceptance. Claude Design governs visual tokens, layouts, component behavior, interaction motion, and responsive intent. Codex may make small technical substitutions only when required for responsiveness or performance and must preserve the design intent.

Claude Design one-shot brief

Design a polished desktop web application named Form Blend + DSPDS. Use a restrained dark 3D training environment and a softer organic analysis layer. Prioritize low cognitive load, spatial clarity, low-latency feel, and progressive disclosure. Produce all screens listed in Section 11, reusable design tokens, component states, loading/recovery/error states, responsive behavior for 1080p and 1440p, and an interactive prototype. Do not invent gameplay or analytics beyond this specification. Active training must contain almost no UI. Results must show plain-language meaning first and expandable evidence second. Deliver implementation notes suitable for Codex.

12. Technical architecture

Technology choices

Layer	Choice	Constraint
Build	Vite + TypeScript	Fast development, typed modules, static deploy.
UI	React	Menus and review only; no per-frame state updates.
3D	Three.js WebGLRenderer	One lightweight scene system; WebGPU is not required for the first proof.
State	Typed finite-state machines and small stores	Assessment and scenario logic must be replayable and testable.
Storage	IndexedDB	Versioned schema, transaction-safe trial commits.
Testing	Vitest + deterministic fixtures; optional Playwright smoke tests	Analysis and state transitions must run without a renderer.
Export	JSON and optional CSV summaries	All output includes schema, app, generator, and analysis versions.

Repository structure

```

form-blend/
  src/
    app/
      App.tsx
      routes.ts
      ui/
      screens/
    engine/
      Renderer.ts
      GameLoop.ts
      CameraRig.ts
      SceneRegistry.ts
    input/
      PointerLockInput.ts
      MouseSampleBuffer.ts
      InputDiagnostics.ts
    profiles/
      schemas.ts
      EquipmentProfileService.ts
    storage/
      db.ts
      migrations.ts
      recovery.ts
      exportImport.ts
    dspds/
      FormalAssessment.ts
      PracticeSession.ts
      scenarios/
        SingleDisplacement.ts
        ChainedFlick.ts
        PredictableTracking.ts
        DynamicTracking.ts
        LineTracing.ts
      analysis/
        vectorMetrics.ts
        gainFit.ts
        lagFit.ts
        confidence.ts
        SpatialModelBuilder.ts
      baseline/
        PhysicalBaseline.ts
        SignatureBuilder.ts
    formBlend/
      FormBlendScenario.ts
      AssistController.ts
      RepAnalyzer.ts
      CoachingRules.ts
    telemetry/
      schemas.ts
      replay.ts
    tests/
  public/
  README.md
  package.json

```

Loop separation

Loop	Responsibility	Rule
Input event path	Timestamp and append raw deltas immediately.	No React state and no per-event object churn after startup.
Simulation	Consume samples, advance scenario FSM, compute displayed camera.	Fixed logical update or timestamp-based integration.
Render	Draw latest scene and HUD state.	May interpolate visuals but never alter raw telemetry.
Analysis worker	Process completed trial arrays and model snapshots.	May run in a Web Worker to avoid blocking interaction.

Loop	Responsibility	Rule
Persistence	Commit completed trial and recovery checkpoint.	Never block the input path.

Core schemas

```

type MouseSample = {
  seq: number;
  tMs: number;
  dx: number;
  dy: number;
  buttons: number;
};

type TrialContext = {
  profileId: string;
  equipmentProfileId: string;
  appVersion: string;
  schemaVersion: number;
  scenarioVersion: string;
  analysisVersion: string;
  seed: string;
  dpi: number | null;
  countsPerCm: number | null;
  selectedCmPer360: number | null;
  measuredCmPer360: number | null;
  verticalFovDeg: number;
  horizontalFovDeg: number;
  viewportPx: [number, number];
  rawInputRequested: boolean;
  rawInputReported: boolean | null;
};

type DspdsTrial = {
  id: string;
  assessmentId: string;
  scenario: "single" | "chain" | "predictableTrack" | "dynamicStatic" |
    "dynamicCamera" | "line";
  formal: boolean;
  intendedAngularPath: AngularSample[];
  targetMetadata: Record<string, unknown>;
  completionMode: "auto" | "click" | "autoWithClickFailsafe";
  completionReason: "settled" | "click" | "timeout" | "technicalInvalid";
  rawSamples: MouseSample[];
  metrics: DspdsTrialMetrics | null;
  valid: boolean;
};

type SpatialModelSnapshot = {
  id: string;
  createdAt: string;
  sourceFormalAssessmentIds: string[];
  latestAssessmentId: string;
  gainByAmplitudeBand: Record<AmplitudeBand, Estimate>;
  gainByDirectionFamily: Record<DirectionFamily, Estimate>;
  reversalAsymmetry: Estimate;
  trackingLagMs: Estimate;
  trackingPhaseError: Estimate;
  jitterAndReversalProfile: Estimate;
  consistency: Estimate;
  sensitivityRange: RangeEstimate | null;
  strengths: EvidenceBackedInsight[];
  weaknesses: EvidenceBackedInsight[];
  confidence: "low" | "moderate" | "high";
};

```

Versioning rule

- Every stored object includes schemaVersion.
- Every scenario records scenarioVersion and seed.
- Every analyzed result records analysisVersion.

- Re-analysis may create a new derived result without modifying the original raw trial.
- Imports must validate known versions and provide a migration preview before changing local data.

Performance rules

- Target stable 144 Hz rendering on the user's desktop-class system with graceful quality reduction.
- No allocations inside the raw event hot path after buffers are created.
- Simple materials, limited bloom, no physics engine, and analytic ray/target intersections.
- Defer chart rendering and heavy analysis until after the active block.
- Record frame and input cadence separately and display reliability warnings when timing quality is inadequate.

13. Codex execution plan and milestones

Implementation order

Step	Deliverable	Completion gate
1	Input capture and diagnostics	Pointer lock, event buffer, gain, Escape recovery, raw/display separation, export fixture.
2	Local profile, storage, autosave, recovery	Trial-level commits survive refresh and import/export round-trip.
3	Sensitivity and FOV setup	Physical rotation remains constant across FOV changes.
4	Optional physical cm/360 tester	Three runs produce stable median, variance, counts/cm, and verification state.
5	DSPDS single-target displacement	Formal and practice modes save valid vectors and block results.
6	DSPDS chained target flicking	Camera auto-centers intended target; chain history and reversal data are correct.
7	DSPDS predictable tracking	Known angular path fits gain and lag from raw input.
8	DSPDS dynamic tracking variants	Static-camera and auto-camera tests remain separate.
9	DSPDS line tracing	Path length, angle, straightness, speed coupling, and lag are reproducible.
10	DSPDS analysis and official model	Snapshots, history, confidence, evidence links, and formal/practice separation work.
11	Four-part physical baseline	Equivalent seeded blocks produce labeled signatures and reliability flags.
12	Form Blend assisted training	Demonstration, hard stop, soft stop, progression, phase validity, and feedback.
13	Session review, trends, and coaching	Useful plain-language insight with expandable evidence and replay.
14	Third-person presentation	Capsule view works without changing angular input or mixing comparisons.

Milestone packaging

Milestone	Steps	Handoff standard
M0 - Foundation	1-4	A stable local app with verified input/profile/view setup.
M1 - DSPDS core gate	5-7	Three core scenarios plus inspectable trial outputs.
M2 - Complete DSPDS	8-10	All five scenarios and official evolving model.
M3 - Physical model	11	Four labeled visible movement signatures.

Milestone	Steps	Handoff standard
M4 - Training proof	12	One polished assisted/unassisted Form Blend scenario.
M5 - Product finish	13-14	Review, trends, coaching, third person, and final UX pass.

One-shot build rule

Limited-engagement optimization

Codex should implement the complete architecture, routes, schemas, scenario interfaces, persistence, tests, and all milestone placeholders in the first pass. It may leave later milestone screens behind explicit feature flags, but must not create a throwaway prototype that requires structural rewrites after DSPDS or profiles are added.

Required commands

```
npm install
npm run dev
npm run build
npm run test
npm run lint

# Optional
npm run test:e2e
npm run export:fixtures
```

14. Acceptance tests and definition of done

Input and view

Test	Pass condition
Event capture	Rapid movement records multiple samples between render frames when browser events provide them.
Pointer lock loss	Pauses safely and never commits a partial trial as valid.
FOV invariance	Raw counts required for a 360 do not change when FOV changes.
Sensitivity	Selected cm/360 produces the expected count-to-angle mapping.
Physical tester	Three entered measurements aggregate correctly and flag high variation.
Raw/display split	Assisted camera behavior cannot modify the raw samples used by analysis.

DSPDS

Test	Pass condition
No input feedback	During a measured response, mouse input never moves the camera, target, reticle, or cursor.
Visible reference	The visual event remains visible; no black-screen memory substitution occurs.
Completion	Automatic, click, and automatic-with-failsafe modes terminate and record correctly.
Formal isolation	Practice data cannot enter an official model snapshot.
Scenario geometry	Intended paths are angular and resolution independent.
Deterministic analysis	Fixture trials produce repeatable gain, lag, path, and confidence outputs.
History	Every formal assessment creates a new immutable record and model snapshot.

Physical baseline and Form Blend

Test	Pass condition
Equivalent targets	All four physical blocks receive matching seed families in shuffled order.
Signature confidence	Overlapping signatures are labeled unresolved rather than forced apart.
Phase order	A user cannot receive a valid repetition by skipping the stop or secondary phase.
Hard stop	Displayed camera freezes while raw carry-through continues to accumulate.
Soft stop	Damping changes only visual gain; raw analysis remains unchanged.
Brute-force resistance	Slow dragging, one ballistic flick, click spam, and sawtooth corrections cannot create a top result.
Feedback	Every rep presents at most one main next-rep cue.

UX and product value

Release gate	Pass condition
Visual quality	Claude Design screens and tokens are implemented consistently at 1080p and 1440p.
Latency feel	No visible UI or analysis task causes input stutter during active movement.
Result credibility	Every conclusion exposes supporting metrics, condition, sample count, and confidence.
Useful coaching	The app turns measured behavior into a specific action rather than only describing data.
Setup burden	A returning user can begin the recommended activity in two primary actions or fewer.
Form gain	After five sessions, the review can compare jitter, correction dependence, blending, stopping, and accuracy against baseline.

Definition of done

- All five DSPDS scenarios exist in formal and practice modes.
- Only formal DSPDS data updates the official evolving model.
- Profiles, equipment contexts, autosave, recovery, and full export/import work.
- The optional cm/360 tester, FOV controls, and first/third-person rules work as specified.
- The four-part physical baseline produces inspectable labeled signatures.
- Form Blend provides demonstration, assisted progression, unassisted training, guardrails, scoring, and coaching.
- The project builds cleanly, passes automated tests, contains no secret keys, and includes a concise README.
- The finished product passes the rejection criteria from the executive summary.

15. Initial tuning and operational protocol

First real-user tuning session

1. Run input diagnostics and export the diagnostic record.
2. Complete the physical cm/360 tester if practical.
3. Complete one formal DSPDS assessment without reviewing trial-level feedback.
4. Inspect scenario agreement, invalid samples, completion-mode behavior, and confidence.
5. Complete the four physical baseline blocks.
6. Complete the 20-repetition Form Blend onboarding.
7. Export the full session package for threshold tuning and defect review.

What may be tuned without architectural change

Tunable family	Examples
Trial timing	Ready cue, minimum movement, settle window, timeout, snap duration.
Target geometry	Angular tiers, radius, wall distance, path speed, chain length.
Confidence thresholds	Minimum samples, dispersion bands, cross-scenario disagreement.
Form Blend thresholds	Gate size, stability speed, hard-stop release, damping zone, validity windows.
Feedback rules	Which measured issue wins the one-cue hierarchy.
Visual quality	Bloom, fog, resolution scale, shadow use, target contrast.

Failure modes and required response

Failure mode	Design response
DSPDS scenarios disagree strongly	Report low confidence and retain scenario-specific estimates; do not average into a false single answer.
Click completion changes movement shape	Compare completion modes; keep automatic-with-failsafe default and warn if click trials differ materially.
Physical signatures overlap	Keep scale and phase coaching; mark anatomical similarity unresolved.
Browser input feels unstable	Stop feature expansion and fix input diagnostics, frame load, and raw/display separation first.
Insights feel arbitrary	Expose evidence, simplify the rule, or remove the insight.
Form Blend improves score but not form	Reweight validity and weakest-phase gates; reduce speed reward.
Onboarding is too long	Make blocks resumable and permit optional tests, but do not hide reliability consequences.

Recommended reassessment rhythm

Do not impose a fixed schedule in version 1. Suggest formal DSPDS reassessment after meaningful training exposure, a profile change, a confidence problem, or a user-requested check. Practice may be repeated freely and never updates the official model.

16. Scope creep with promising ROI

No expansion should begin until input quality, DSPDS core validity, profile persistence, and one complete Form Blend scenario are proven. The following additions have credible value but increase time or testing cost.

Priority	Feature	Potential ROI	Cost
1	Blinded closed-loop sensitivity validation	Tests whether DSPDS intuitive ranges transfer to visible accuracy, stopping, smoothness, and correction behavior.	Medium
2	Personal scenario generator from DSPDS weaknesses	Automatically creates targeted direction/amplitude/reversal blocks while preserving balanced exposure.	Medium
3	Rich replay and path geometry	Makes carry-through, first arrival, correction tails, and spatial-model differences immediately understandable.	Medium
4	Native desktop wrapper or HID companion	Improves fullscreen behavior, input diagnostics, and browser ambiguity.	Medium-high
5	Game sensitivity and FOV profiles	Improves transfer and comparison across games without changing core analysis.	Medium
6	Optional webcam/IMU anatomy validation	Could test whether arm/wrist/finger labels match observed motion.	High

Priority	Feature	Potential ROI	Cost
7	Curved and depth-varying DSPDS paths	Extends the internal model to parallax, depth, and non-linear trajectories.	Medium-high
8	Animated third-person body model	Potential teaching visual, but little measurement value without body tracking.	High / low initial ROI

Best first expansion

Closed-loop sensitivity validation has the highest immediate scientific and training ROI because it compares what feels spatially natural with what actually produces good visible aim. It should be the first follow-up milestone after the initial release.

Appendix A. Initial configuration defaults

```
export const DEFAULTS = {
  input: {
    dpi: 1600,
    cmPer360: 40,
    verticalFovDeg: 90,
    viewMode: "firstPerson",
    dspdsCompletionMode: "autoWithClickFailsafe",
  },
  dspds: {
    formalTargetMinutes: 11,
    settleWindowMs: 150,
    completionTimeoutMs: 3000,
    singleAnglesDeg: [2, 4, 8, 12, 18, 27, 40],
    chainLength: [5, 8],
    cameraSnapMs: 100,
    predictableCycles: 2.5,
    currentModelSnapshotCount: 3,
    recencyWeights: [1.0, 1.5, 2.0],
  },
  baseline: {
    armReps: 16,
    wristReps: 16,
    fingertipReps: 20,
    freestyleReps: 24,
  },
  formBlend: {
    onboardingMaxReps: 20,
    assistedStages: ["demoHard", "hard", "reducedHard", "soft", "cue", "none"],
  },
  storage: {
    rawSessionRetentionMinimum: 100,
  },
};
```

All defaults are starting values. They must remain centralized, typed, exported, and visible in developer diagnostics so the first real session can tune them without rewriting scenario logic.

Appendix B. One-shot Codex directive

Build the complete Form Blend + DSPDS desktop browser application described in this specification. Use Vite, React, TypeScript, Three.js, IndexedDB, Vitest, and a clean separation between raw input, simulation, rendering, analysis, and persistence.

Implement the full repository structure, typed schemas, migrations, autosave/recovery, JSON import/export, diagnostics, sensitivity/FOV setup, optional physical cm/360 tester, all five DSPDS scenarios in formal and practice modes, official model snapshots updated only by formal assessments, the four-part physical baseline, and one complete assisted Form Blend training scenario.

Do not substitute opaque AI analysis, fake telemetry, or random scores. Every insight must be derived from stored samples and named metrics. Do not let React own the real-time loop. Do not modify raw samples to match assisted camera behavior. Do not add accounts, servers, multiplayer, achievements, or unrelated game systems.

Implement Claude Design artifacts as the visual authority when supplied. Until then, build the screens and component contracts with a restrained polished dark interface that can be restyled without altering scenario logic.

Create deterministic fixtures and tests for vector projection, implied sensitivity, lag/gain fitting, line metrics, confidence, formal/practice separation, profile migration, Form Blend state transitions, and weakest-phase validity. Include a README with setup, controls, data locations, export format, diagnostics, known browser limitations, and tuning defaults.

The app must run with `npm install`, `npm run dev`, `npm run build`, `npm run test`, and `npm run lint`.

Appendix C. Compact implementation checklist

Category	Must exist before handoff
Repository	Typed source, scripts, tests, README, no secrets, clean build/lint.
Input	Pointer lock, unadjusted request, ring buffer, diagnostics, pause recovery, raw/display split.
Profile	One user, multiple equipment contexts, IndexedDB migration, autosave, recovery, export/import.
View setup	DPI, selected/measured cm/360, optional physical tester, FOV conversion, first/third-person rules.
DSPDS governance	Formal/practice separation, immutable formal history, official snapshots from formal data only.
DSPDS scenarios	Single, chain, predictable tracking, dynamic static/camera, line tracing.
DSPDS analysis	Gain, direction, lag, phase, drift, path, consistency, confidence, sensitivity range, evidence links.
Physical baseline	Arm, wrist, fingertip, freestyle blocks with equivalent seeds and labeled signatures.
Form Blend	Demonstration, hard/soft stop, progression, guardrails, weakest-phase result, one-cue coaching.
UI/UX	Claude Design coverage, polished active HUD, results, profile, history, failure/recovery states.
Acceptance	No lag, arbitrary results, cheap UX, excessive setup, or data without coaching/form benefit.

Final product statement

Build a local 3D training tool that first measures how the user naturally translates visual space into physical mouse movement, then measures the user's labeled physical movement signatures, and finally teaches a controlled blended aiming sequence. The output must be useful, inspectable, visually polished, and oriented toward real-game improvement rather than score farming.